

Minimum induced drag of superelliptical closed lifting systems with a loaded central body

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Abstract

Closed or “box” wings have been recognized since Ludwig Prandtl’s foundational work to reduce induced drag compared to flat, planar wings of the same total wingspan. However, when trying to translate these theoretical concepts into practical ring-wing aircraft designs, two critical questions emerge that classical literature leaves unanswered: (i) how do smoothly curved, closed wing shapes compare to a sharp-cornered rectangular box wing at identical height-to-span ratios (h/b)? and (ii) what is the aerodynamic cost of placing a central cabin body inside the ring and requiring it to carry a share of the total lift? Using a minimum-induced-drag formulation in the Trefftz plane, this study addresses both questions. The continuous wake trace is discretized with piecewise-constant circulation panels and solved as an equality-constrained quadratic program. The optimal span efficiency of an elliptical ring is found to follow a simple relation, $e = 1 + h/b$, with Richardson extrapolation of the observed first-order grid convergence reducing the residual to 10^{-6} , strong evidence of an exact law. A superellipse sweep shows a $p = 4$ “squircle” captures roughly three quarters of the box-wing benefit, and a finite-chord vortex-lattice cross-check confirms these ideal efficiencies are attainable on real surfaces, with untwisted rings reaching 98.6–100% of the ideal. A combined design point—a $p = 4$ superelliptical ring of $h/b = 0.5$ with a low-set cabin carrying 15% of the lift—achieves $e = 1.61$, a 38% reduction in induced drag relative to an ideal planar wing of equal span.

Summary of key findings

- **Validation.** The numerical tool was validated against three classic benchmark cases: the standard planar wing ($e = 1.00$), Prandtl’s rectangular box wing ($e \approx 1.46$ at $h/b = 0.2$), and the circular ring wing ($e = 2.00$).
- **The $e = 1 + h/b$ law.** Across the entire tested height-to-span range, the optimal span efficiency e of an elliptical ring follows the remarkably simple relation $e = 1 + h/b$. Richardson extrapolation of the first-order grid convergence reduces the residual error to 10^{-6} , providing strong evidence that this is an exact theoretical law.
- **Squaring the ring.** Mapping the superellipse family between a pure ellipse ($p = 2$) and a rectangular box ($p \rightarrow \infty$) reveals rapid performance gains: a $p = 4$ “squircle” captures roughly 75% of the ideal box-wing efficiency benefit, and $p = 6$ captures 83–85%.
- **Central-cabin loading penalty.** Asking a central cabin to carry lift incurs an aerodynamic penalty, smallest when the body blends into either the top or bottom arc, and worst when suspended at mid-height.

- **Practical attainability.** A finite-chord vortex-lattice method confirms these ideal efficiencies are realizable on physical surfaces: untwisted rings at uniform incidence achieve 98.6–100% of the Trefftz ideal, and less than 1° of twist recovers the remainder.
- **Combined design point.** A $p = 4$ superelliptical ring ($h/b = 0.5$) with a low-set cabin carrying 15% of the lift achieves $e = 1.61$ —a 38% reduction in induced drag versus an ideal planar wing of equal span.

1 Introduction

Induced drag—the resistance created as a byproduct of generating lift—accounts for approximately one-third of a commercial transport aircraft’s drag during cruise, and nearly half during climb. In 1921, Max Munk proved that for a conventional planar (flat) wing, the minimum induced drag occurs under an elliptical lift distribution, giving the classic induced-drag equation

$$D_i = \frac{L^2}{\pi q b^2 e}, \quad (1)$$

where $e = 1$ represents the maximum efficiency of a standard flat wing.

Nonplanar and closed lifting systems can exceed $e = 1$ (Table 1). Real-world ring-wing aircraft concepts deviate from ideal closed shapes in two ways. First, in *structural and aerodynamic geometry*: real wings require smooth curves rather than sharp right angles to reduce structural weight, avoid stress concentrations, and minimize corner interference drag, yet the performance of intermediate shapes between a pure ellipse and a rigid box has remained largely unexplored. Second, in *integrated fuselage loading*: practical designs typically nest the passenger cabin inside the ring, which in the Trefftz plane acts as a detached central lifting surface, and the efficiency cost of forcing this cabin to carry a portion of the aircraft’s weight has not been systematically quantified. This paper addresses both gaps using a validated numerical framework.

Table 1: Closed and nonplanar lifting systems and their span efficiency.

Wing geometry	Historical origin	Performance characteristic
Planar wing	Munk (1921)	Baseline span efficiency, $e = 1.00$
Box wing	Prandtl (1924)	“Best wing system”; $e \approx 1.46$ at $h/b = 0.2$
Annular / ring wing	Cone (1962)	Closed circular system, $e = 2.00$

2 Methodology

2.1 Trefftz-plane formulation

Far downstream of a lifting aircraft, the trailing wake flattens into a two-dimensional vortex sheet. Its cross-section trace C in the transverse plane (the Trefftz plane) completely dictates the overall induced drag. For a circulation distribution $\Gamma(s)$ along arc length s ,

$$L = \rho V \int_C \Gamma \, dy, \quad D_i = \frac{\rho}{2} \int_C \Gamma w_n \, ds, \quad (2)$$

where w_n is the normal downwash velocity induced in the Trefftz plane by the shed vorticity $-\mathrm{d}\Gamma/\mathrm{d}s$. Minimizing D_i for a fixed target lift L yields the optimal wing loading and determines the span efficiency e .

2.2 Numerical discretization

The lifting trace is split into n panels, each carrying a piecewise-constant circulation Γ_i . Each panel sheds two-dimensional point vortices at its endpoints; at adjacent panel boundaries these merge into a physical shed-vortex strength $\Gamma_i - \Gamma_{i+1}$. Control points are situated at panel midpoints. Substituting these discrete elements into the integral equations converts the problem into matrix form,

$$D_i = \mathbf{\Gamma}^\top A \mathbf{\Gamma}, \quad L = \mathbf{b}^\top \mathbf{\Gamma}. \quad (3)$$

The total-lift requirement (and any individual component-lift constraints) are collected into $B^\top \mathbf{\Gamma} = \mathbf{d}$, and the global optimum is computed directly by solving the Karush–Kuhn–Tucker (KKT) linear system

$$\begin{pmatrix} 2A & -B \\ B^\top & 0 \end{pmatrix} \begin{pmatrix} \mathbf{\Gamma} \\ \boldsymbol{\lambda} \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ \mathbf{d} \end{pmatrix}. \quad (4)$$

Closed geometries are resampled with uniform arc-length spacing to prevent numerical ill-conditioning near corners; open traces use standard cosine spacing. Resolutions up to $n = 1536$ panels were evaluated.

3 Tool validation and numerical convergence

The numerical formulation was validated against three classical analytical targets (Table 2).

Table 2: Validation against classical results.

Case study	Grid resolution	Computed e	Classical target	Source
Planar wing	$n = 400$	1.0031	1.0000	Munk (1921)
Box wing ($h/b = 0.2$)	$n_{\text{span}} = 64 \rightarrow 512$	1.4721	≈ 1.46	Prandtl (1924)
Circular ring	$n = 128 \rightarrow 1024$	2.0000	2.0000	Cone (1962)

Grid convergence. Kernel discretization error behaves strictly as a first-order system. For a planar wing, the bias $e - 1$ halves smoothly with every doubling of panel density, falling from $+6.3 \times 10^{-3}$ at $n = 200$ to $+3.9 \times 10^{-4}$ at $n = 3200$.

Corner stability. Resolving sharp corners at high superellipse exponents ($p = 12$) showed minimal grid sensitivity: efficiency shifted by only 1.6×10^{-4} when scaling resolution from $n = 384$ to $n = 3072$.

4 Results and analysis

4.1 The elliptical ring law: $e = 1 + h/b$

Comparing rectangular box wings against elliptical rings across height-to-span ratios from 0.05 to 1.0 confirms that the box wing is superior at every height. For instance, at $h/b = 0.3$ the rectangular box reaches $e = 1.658$, whereas the pure ellipse reaches $e = 1.302$. However, the elliptical ring data reveals a remarkably clean pattern (Fig. 1): across all tested heights the calculated efficiency follows an exact linear relationship,

$$e_{\text{ellipse}} = 1 + \frac{h}{b}. \quad (5)$$

At $n = 1536$ the maximum numerical deviation from this equation is under 7.9×10^{-4} . Applying Richardson extrapolation to the grid convergence sequence drives the residual down to 1×10^{-6} . This provides strong evidence that $e = 1 + h/b$ represents an exact law for elliptical closed traces. The author has not located this relation stated in the accessible literature; it may nonetheless be implicit in closed-form analyses of circular annular wings (Cone, 1962; Demasi, 2007), and an analytical derivation is left as future work.

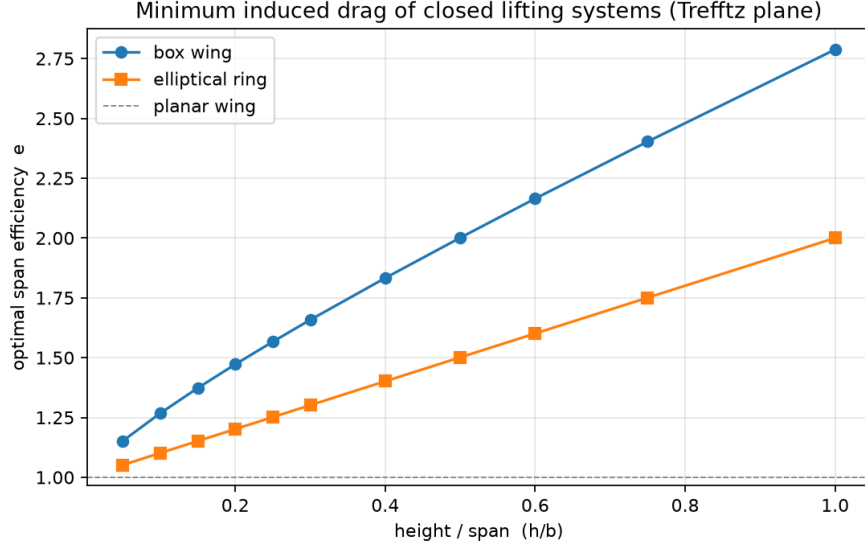


Figure 1: Optimal span efficiency of the box wing and the elliptical ring versus height-to-span ratio. The elliptical ring follows $e = 1 + h/b$.

4.2 Squaring the ring: superellipse parameter sweep

To evaluate practical shapes between a pure ellipse and a rectangular box, we analyze the superellipse family

$$\left|\frac{y}{a}\right|^p + \left|\frac{z}{c}\right|^p = 1, \quad (6)$$

where $p = 2$ is a standard ellipse and $p \rightarrow \infty$ approaches a rectangle. Measuring the fraction of the box wing’s efficiency advantage gained by each shape—the *benefit capture*, defined as $(e - 1)/(e_{\text{box}} - 1)$ —reveals that aerodynamic performance saturates rapidly with increasing p (Table 3, Fig. 2).

Table 3: Benefit capture of the superellipse family.

Exponent p	Geometry	Capture ($h/b = 0.3$)	Capture ($h/b = 0.5$)
$p = 2$	Pure ellipse	0% (baseline)	0% (baseline)
$p = 3$	Slightly rounded	63%	67%
$p = 4$	“Squircle”	73%	76%
$p = 6$	Strongly squared	83%	85%
$p = 12$	Sharp superellipse	93%	94%

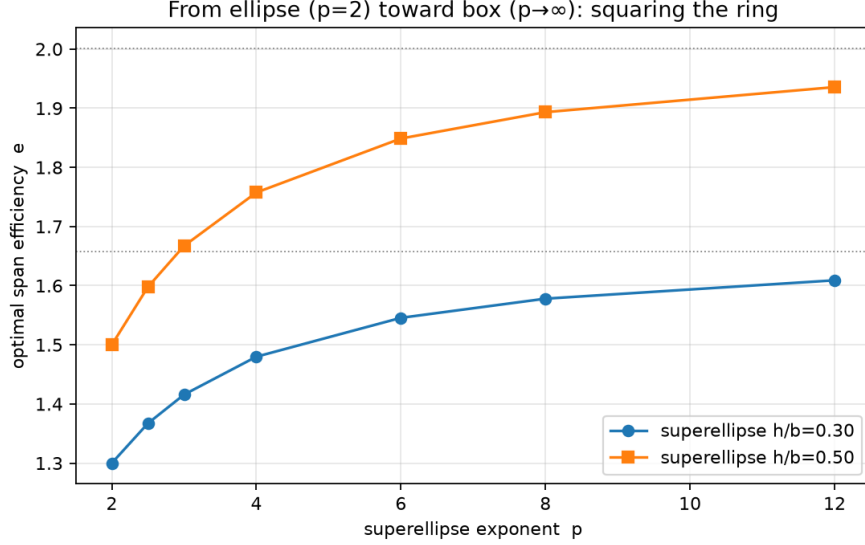


Figure 2: Span efficiency versus superellipse exponent p at two height-to-span ratios. Dotted lines mark the box-wing values ($p \rightarrow \infty$).

Design takeaway. Sharp 90° corners are unnecessary. A gently rounded $p = 4$ or $p = 6$ squircle captures 75–85% of a box wing’s drag reduction while eliminating the severe corner stress concentrations and high viscous interference drag associated with a true box.

4.3 Aerodynamic penalty of central-cabin loading

When a central fuselage/cabin is placed inside the ring and required to generate a fraction f of total lift ($L_{\text{body}} = f L_{\text{total}}$), efficiency drops.

Lift-share impact. For a mid-height cabin ($0.5b$ wide), induced drag increases quadratically with f . At $h/b = 0.3$, increasing body lift share from $f = 0$ to $f = 0.20$ drops span efficiency from $e = 1.302$ to $e = 1.143$ (Fig. 3); beyond $f \approx 0.30$ the ring wing performs worse than a standard flat wing.

Vertical-placement impact. Holding lift share constant at $f = 0.15$, vertical position plays a crucial role (Fig. 4). Mid-height ($z/h = 0.50$) is the worst placement ($e = 1.145$): the cabin’s shed vortices are maximally exposed within the open interior of the ring. Arc-hugging ($z/h = 0.10$ or 0.90) is best ($e = 1.226$): the body’s trailing wake partially merges with the ring’s natural wake sheet.

Body-width sensitivity. Widening the lifting body dramatically reduces the penalty. At mid-height with $f = 0.20$, a narrow cabin ($0.3b$ wide) degrades efficiency to $e = 0.88$, whereas a wide lifting body ($0.7b$ wide) maintains $e = 1.23$.

Two golden rules for fuselage integration follow: (1) *blend into the arcs*—integrate the cabin directly into the upper or lower arc of the ring, never suspended at mid-height; and (2) *make it wide*—if the cabin must lift, use a broad lifting-body shape to distribute shed vorticity over a larger spanwise extent.

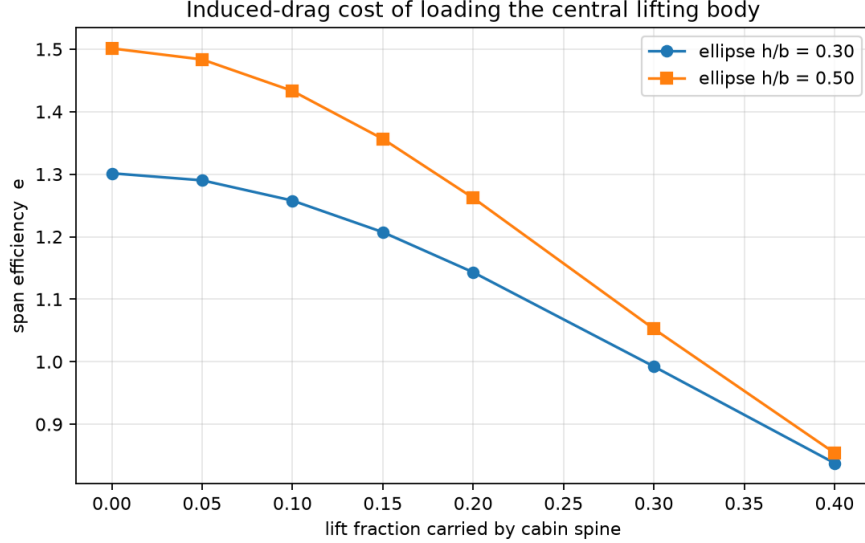


Figure 3: Efficiency versus lift fraction carried by a mid-height central body, for two ring heights.

4.4 Finite-chord validation (vortex-lattice check)

To ensure these Trefftz-plane limits are achievable on physical surfaces with real chords, a horseshoe vortex-lattice method (VLM) was constructed using matching geometry traces (144 circumferential strips, 8 chordwise panels, chord-to-span ratio $c/b = 0.10$). The results (Table 4, Fig. 5) demonstrate that closed ring wings are essentially self-optimizing: at a constant angle of attack with zero wash-in or wash-out, ring wings spontaneously achieve 98.6% to 100% of their maximum theoretical efficiency, and applying less than 1° of geometric twist completely recovers the remaining 1–2%.

Table 4: Finite-chord VLM versus Trefftz ideal (144 strips, $c/b = 0.10$).

Geometry	h/b	Ideal e	Untwisted e	% ideal	Optimised e	Max twist
Circular ring	1.0	1.9995	1.9995	100.0%	1.9995	0.00°
Elliptical ring	0.5	1.5061	1.4851	98.6%	1.5061	0.76°
Squire ($p = 4$)	0.5	1.7569	1.7413	99.1%	1.7569	0.63°

5 Combined design point

Combining these findings into a unified, practical transport-aircraft layout gives a superellipse exponent $p = 4$ (“squire”), height-to-span ratio $h/b = 0.5$, and a low-set cabin ($z/h = 0.15$, width $0.5b$) carrying 15% of total aircraft lift. Despite accounting for the aerodynamic penalty of a cabin carrying 15% of the flight load, this integrated configuration achieves $e = 1.61$ (Fig. 6)—a 38% net reduction in induced drag compared to an ideal flat wing of equal total span.

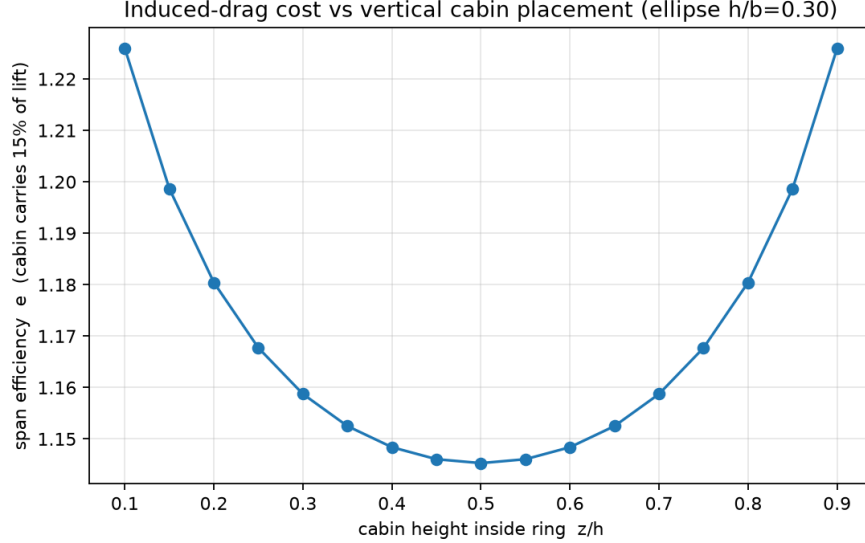


Figure 4: Efficiency versus vertical position of the central body at a fixed 15% lift share (ellipse, $h/b = 0.3$). Mid-height is the worst placement.

6 Preliminary experimental validation

[This section is left for the author to write, describing the subscale cardboard-glider flight test in which the vortex-lattice neutral-point prediction (balance point 7.5–8 cm behind the bottom wing’s leading edge) was confirmed in flight, and a glide ratio of roughly 3.3–3.8 was measured. See the bullet notes in the source file.]

7 Limitations

[This section is left for the author to write, stating the study’s limitations honestly: the analysis is inviscid (induced drag only), assumes ideal loading, is untrimmed, and has not been validated in viscous flow beyond the subscale glider; the $e = 1 + h/b$ relation is numerical evidence, not an analytical proof. See the bullet notes in the source file.]

8 Summary of engineering rules

1. **Use squircles, not boxes.** Adopt $p = 4$ to $p = 6$ superellipses to obtain 75–85% of a rectangular box wing’s efficiency advantage while eliminating corner structural weight penalties and severe junction drag.
2. **Never suspend cabins at mid-height.** Attach lifting bodies directly to the top or bottom arc of the ring so the wake sheets merge smoothly.
3. **Keep cabin lift shares moderate.** Limit internal-body lift contributions to $\lesssim 15\%$ of total aircraft weight, and maximize body width where possible.
4. **Rely on natural downwash.** Untwisted, constant-incidence closed rings inherently achieve $\geq 98.6\%$ of their ideal theoretical span efficiency without complex wash-out schedules.

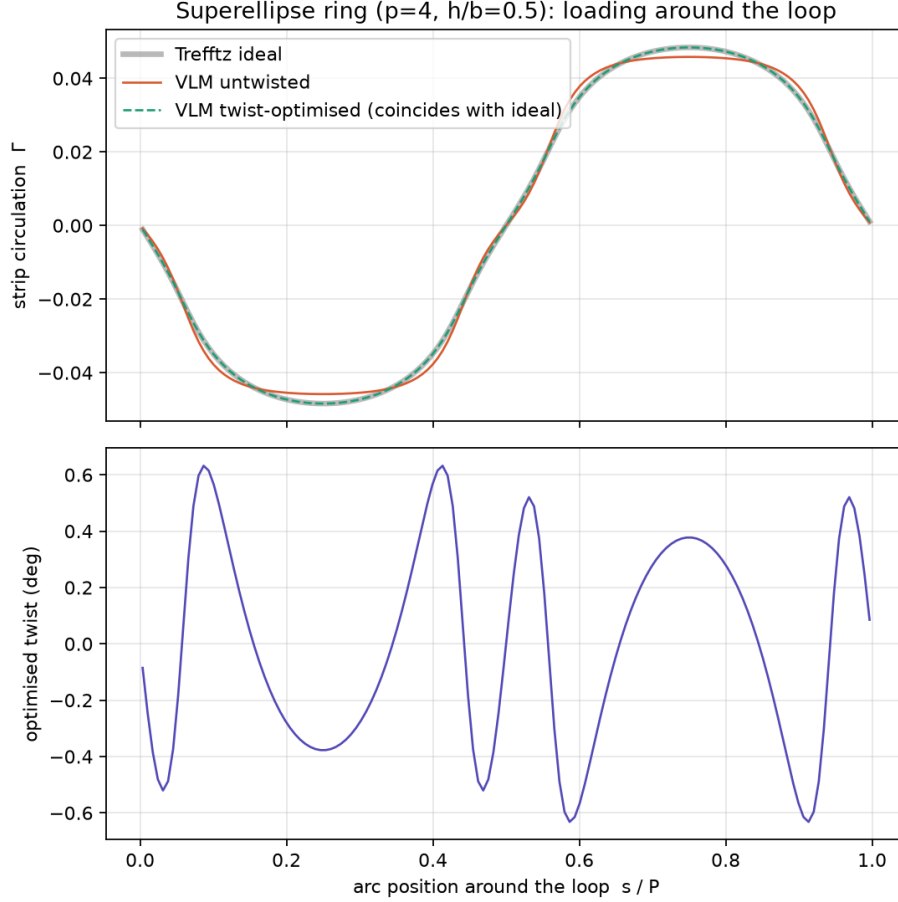


Figure 5: Superellipse ring ($p = 4$, $h/b = 0.5$): strip circulation around the loop for the Trefftz ideal, the untwisted VLM, and the twist-optimised VLM (top), and the sub-degree optimised twist schedule (bottom).

Data and code availability

The Trefftz-plane solver, the vortex-lattice cross-check, all run scripts, data tables, and the geometry generator are available at <https://github.com/SomeshParganiha/ring-wing-paper> and archived on Zenodo, doi:10.5281/zenodo.21205563.

Acknowledgements and disclosure

This manuscript was written by the author. The author conceived the study, learned the underlying aerodynamic theory, directed and reviewed the analysis, reproduced the results, and built and flight-tested the subscale demonstrator. The numerical tools used to produce the results—the Trefftz-plane optimiser, the finite-chord vortex-lattice cross-check, and the geometry/STL generator—were implemented with the assistance of an AI system (Anthropic Claude) working under the author’s direction.

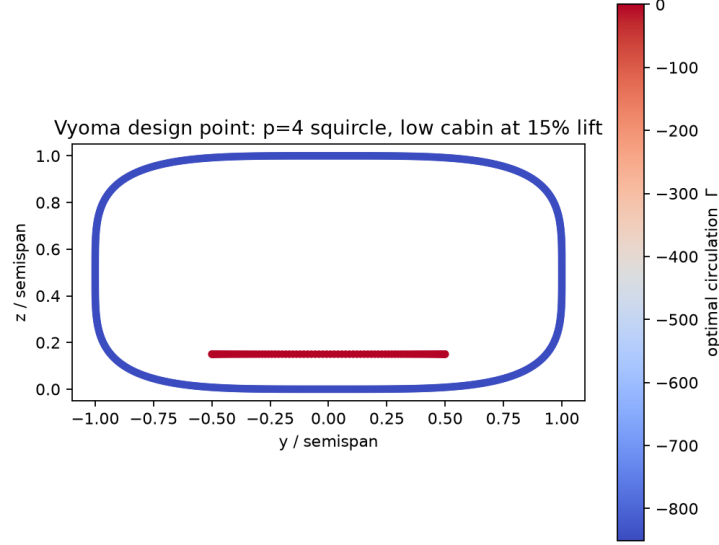


Figure 6: Optimal circulation at the combined design point: $p = 4$ superellipse, $h/b = 0.5$, low-set central body at 15% lift share.

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